Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

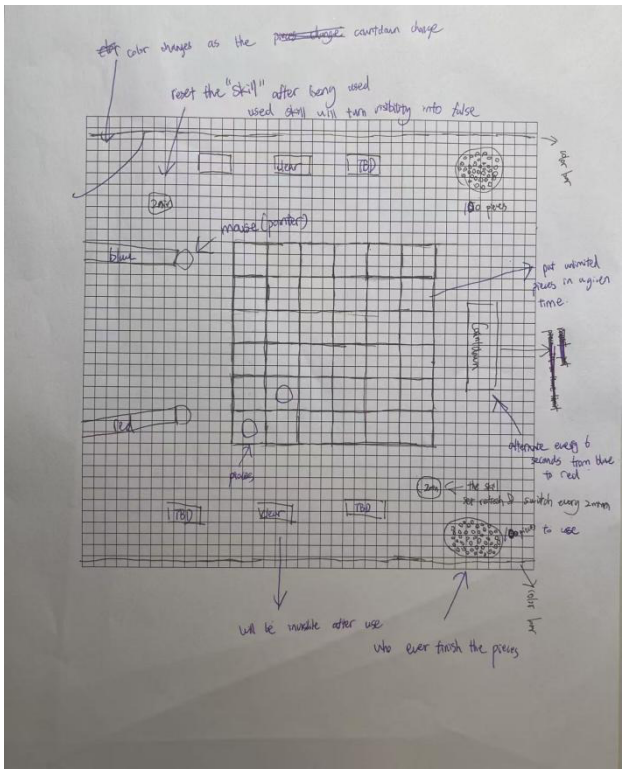
Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

This program is a game where players need to put as much pieces as possible to win against the opponent. Players will alternate to put unlimited pieces every 6 seconds (onStep) and try to put as much as they can to spend all their 100 pieces to win. There will be an additional color bar on the top left to show players whose turn it is. They would also have a “skill” that can clear all their added pieces (clear group) to create space to put more pieces. This skill can only be used once every 12 second. There is two ways to win in this game. The first is to use all your 100 pieces before your opponent do. The second is when the time runs up, you have used more pieces than the opponent. (Yes, there will be a big timer). This is a game that tests players’ strategy, timing, and hand speed.

Canvas Design

Sketch your canvas. Label any animation or user interaction.



Code Design

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Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
onStep ()	I need to use this to code the countdown for the entire game, when the players switch every 6 seconds, and the countdown to refresh the clear "skill"
onMouseMove (x,y)	I need to use this to code the mouse for the two players, as player blue would have a mouse with blue border and player red would have a mouse with red border.
onMousePress (mouseX, mouseY)	I need to use this to allow players to put pieces to their desired spots within a particular domain and range. This press can also allow players to use the "skill" when they press on them.
group.clear()	This is basically the core idea of the "skill", where players can use this to clear their own added pieces to create more space to put more pieces.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

One addition to this project was the additional way of winning. I originally planned to design only one way of winning, which is to use up all the 100 pieces. I believe 100 pieces would make players feel more enjoyable to play this game as they will have more room for mistakes. However, I found out that it might be boring sometimes to play until 100 pieces. Thus, I believe adding a time restrain to this game would be an good idea to both prevent tediousness and push the pace of the game. One aspect I had to

incorporate to this idea is the comparison of the pieces they left in the “piecebox”, which I compare the pieces number to determine the winner after the time is up. I had to adjust only a little to incorporate this new feature, which makes this feature a nice addition to the mechanism of this game. Since the time countdown can be easily added to the onStep function, while the winning condition can all be added in the end.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
onStep()	#71-75	This is a function called automatically and repeatedly as time passes	The purpose of this code is to refresh the clear “skill” every 12 seconds. Clearly, the countdown will decrease 10 values per second until it reaches zero, which the skill will be visible and usable by then
onMouseMove (x,y)	#97-108	This is a function that is called whenever the mouse is moved	The purpose of this code is to show which player’s turn it is, where different wrist from different limb will become the pointer as color alternates. Part of the reason for this code is for overall aesthetic for the game.
group.add()	#110 & 129-134	This is a group method used to add new shape to a certain group	The purpose of this code is for players to put pieces within the few given coordinates in the boxes. Every new circle being added is a new piece being added, which the piece number will decrease by one.
group.clear()	#145-152	This is a group method used to remove all the shapes at once from a group	The purpose of this code is to demonstrate the core idea of my clear “skill”. As it is pressed, the pieces from the player’s group will be removed, which creates more spaces to put pieces. The skill will also become invisible until it is refreshed.

Future goals

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Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

If I had more time to start from scratch, I would try to alternate the mechanism of how this game is played and how players win in this game. Specifically, I would like to determine the winner by determining how many boxes they occupy using their pieces after the time runs out or both pieces are depleted. Additionally, I would also like to add more skills like the clear “skill” and even randomize which one will become visible whenever it refreshes. This is actually my original plan, but I didn’t stick to this plan since it requires too much skills beyond this unit and what we’ve practiced. I believe the product I have right now, which were my changes, is beneficial because I am able to use the exact skills I learned from this unit and practice them. My approach could’ve been improved by outlining the specific concept I can use for each mechanism of the game earlier and discuss them with the teacher. I believe I did make a good use of the resources provided by the teacher and opinion given by my peers, since I can actually build a systemic game that works (unlike last time). I am also able to fit the skills we learned to class as perfect function to my game, which I was pretty proud of. Some additional resources could be tutorials that teach me how to creatively use all the concepts we have learned. I believe that I didn’t reach the very best of combining these concepts all together with mathematics to achieve the function I intend to achieve. These tutorials can be applied to improve my time management skill, which I can start planning on watching these videos along with reviewing the concept and coding the game in certain times of a week. This can certainly lead to a smoother process to finish this coding